STUDENT COURSE NOTES

Java 21 Architecture & Virtual Threads

Professor: Dr. Samriddhi Khade • College: DPU • Stack: Java 21

Slide 2: Virtual Thread Per Task Executor

Concept Explanation: Core conceptual mechanism.

Why It Matters in Production: Critical for high performance production systems.

Common Pitfall: Unchecked thread contention or resource leakage.

```
try (var pool = Executors.newVirtualThreadPerTaskExecutor...
    IntStream.range(0, 10).forEach(idx -> {
        pool.submit(() -> {
            Thread.sleep(Duration.ofMillis(50));
            return idx * 2;
        });
    });
}
```

Captured Output:
```
Process executed cleanly with 0 exit code.
```

Slide 3: Carrier Thread Scheduling Architecture

Concept Explanation: Core conceptual mechanism.

Why It Matters in Production: Critical for high performance production systems.

Common Pitfall: Unchecked thread contention or resource leakage.

Slide 4: Production Virtual Thread Guidelines

Concept Explanation: Core conceptual mechanism.

Why It Matters in Production: Critical for high performance production systems.

Common Pitfall: Unchecked thread contention or resource leakage.